Ch 6.4: Curse of Dimensionality

Lecture 19 - CMSE 381

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Announcements

Last time:

- PCA/PCR

This lecture:

- 6.3: PLS
- 6.4: Issues with higher dimensions

Announcements:

- Homework due Friday
- Monday is review day
- Wednesday is Exam
  - 8.5" x 11" cheat sheet
  - Basic calculator
  - Covers content since last exam (Chapters 5 and 6)

Section 1

Last time

Shrinkage

Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to:

Least Squares:

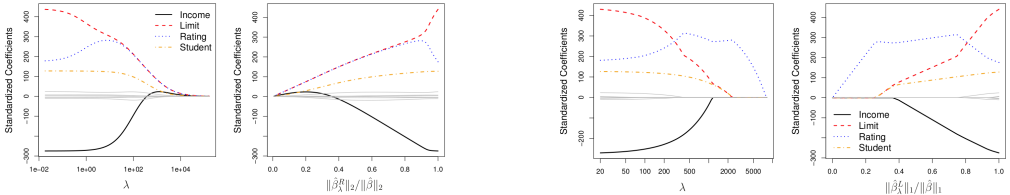
No constraints

Ridge:

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$



Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$M \ll P$

$$Z_1, \cdots, \cancel{Z_M}\ Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

The goal:

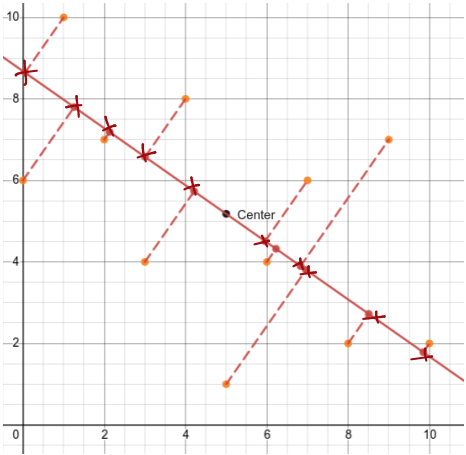
- Find good φ's
- Fit regression model on Z_i's using least squares

coefficients to learn (prev. β)

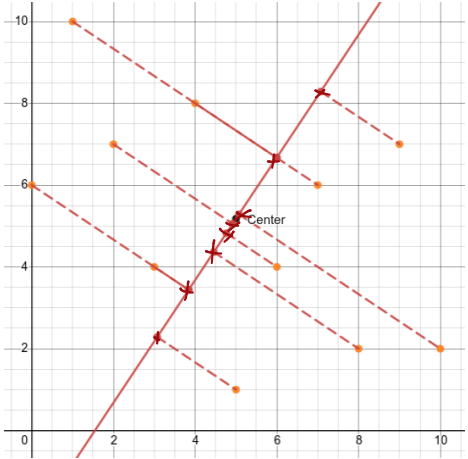
$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

- Hope that lower dimensions means less overfitting
- Remember that interpretation not the same as shrinkage/subset selection of variables

PCA - First PC

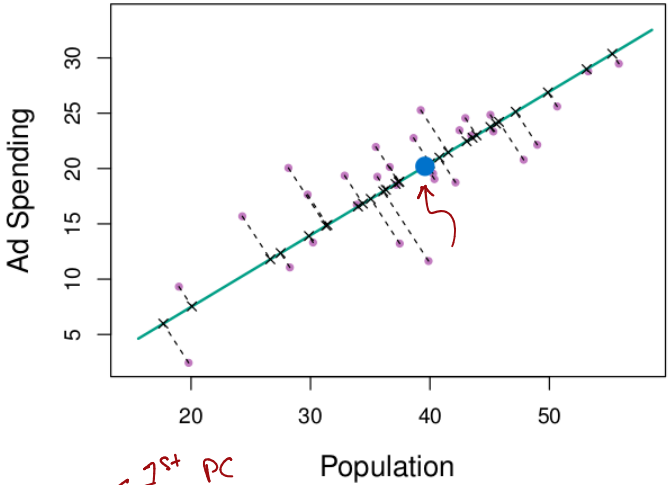


Projection has high variance



Projection has low(er) variance

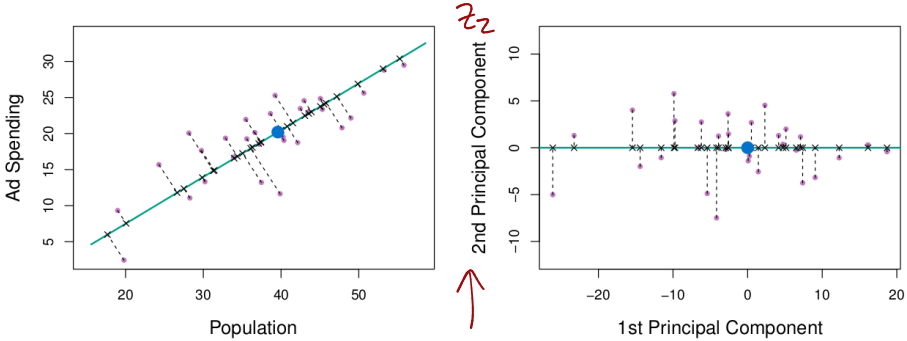
Projection onto first PC



1st PC

$$Z_1 = 0.839 \cdot (\text{pop} - \overline{\text{pop}}) + 0.544 \cdot (\text{ad} - \overline{\text{ad}})$$

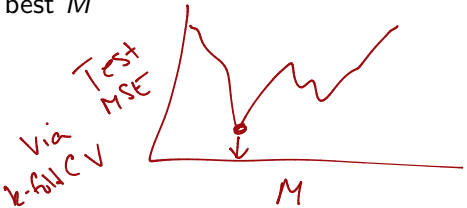
Drawing points in PC space



$p = 2$ dimensions

Principal Components Regression (PCR)

- Take new features Z_i
- Run regression
- Maybe do CV for a bunch of choices of M (where $M =$ number of Z_i features used) to pick a best M



- You can't directly use the coeff. to interpret the original inputs
- Each PC is a linear combo of the input variables so potentially all X_i's are involved
  ⇝ Not subset selection!

Figuring out the original model from the PC model

- We have three input variables X_1, X_2, X_3
- We have two PCs,

$$Z_1 = 2(0.266 \cdot X_1 - 0.077 \cdot X_2 + 0.961 \cdot X_3)$$

$$Z_2 = -4(0.968 \cdot X_1 + 0.136 \cdot X_2 - 0.254 \cdot X_3)$$

- Using linear regression, we learn the model

$$Y = -3 + 2Z_1 - 4Z_2.$$

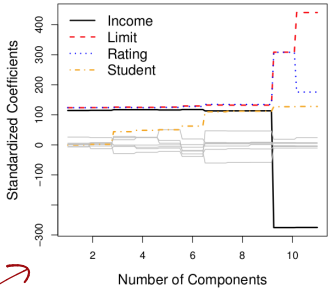
- What are the coefficients for the model in terms of the X_i's?

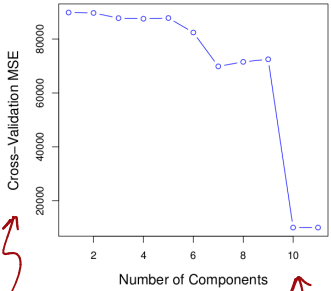
$$Y = -3 + 2Z_1 - 4Z_2$$

$$= -3 + (2 \cdot 0.266 - 4(0.968)) \cdot X_1 + (\quad) X_2 + (\quad) X_3$$

$$= -3 - 3.34 X_1 - 0.698 X_2 + 2.938 X_3$$

Example on Credit dataset





~ Data has $p = 11$

Test MSE approx

M

Want $M = 10$

- learned coefficients on the X_i's
- Behaves a bit like ridge / lasso

Section 2

Partial Least Squares (PLS)

Supervised alternative

PCR: Non-supervised

- No input from the Y values when doing PCA to learn the PC's
- No guarantee the-1 the chosen directions help predict Y.

Partial Least Squares (PLS):

- Identify new features $Z_1, \cdots, Z_M$ linear combos of original where quality measure involves Y
- Fit linear model using least squares on these M features

First direction Z_1 for Partial Least Squares (PLS)

For each input $X_1, \ldots, X_p$

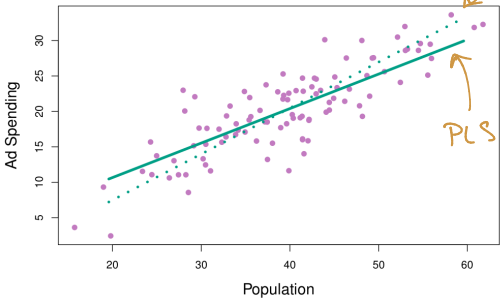
$y = \beta_0 + \beta_1 X_j$

- Set φ_{j1} equal to the coefficient from simple linear regression of Y onto X_j
- The first direction is

$$Z_1 = \sum_{j=1}^{p} \varphi_{j1} X_j$$

• PLS puts the highest weight on variables most associated with the response Y

PCA: 1st PC



Ex. Prediction of Y =Sales (not shown) on X_1 =Population and X_2 =Ad Spending

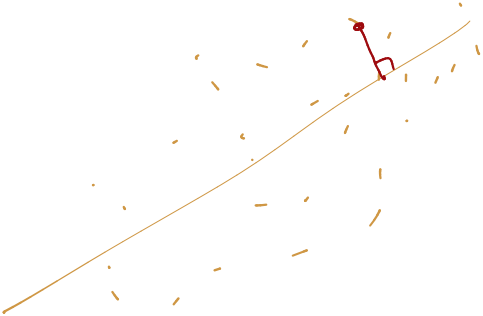
- Solid green: First PLS direction
- Dashed: First PC direction

• PLS chose line less change in the ad direction; PLS is better predictor

Second (and more) PLS directions

- Regress each variable on Z_1 and take residuals
- Compute Z_2 using *orthogonalized data* same as for Z_1

- M can picked using cv techniques

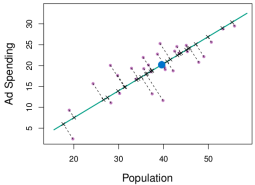


Code example on hitters data

PCA vs PLS

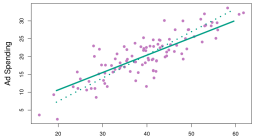
PCA /PCR

- Unsupervised dimensionality reduction
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



PLS

- Supervised dimensionality reduction
- Choose component Z_1 by using simple regression coefficients of each X_i onto Y
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained
- Not a particular benefit, so usually default to PCA unless you have a good reason for this



Section 3

Issues in Higher Dimensions

High-Dimensional Data

n = # data pts p = # input variables

Low-Dimensions

$n \gg p$

- Can do any tool used in class so far
- Historically, most problems were here.
- Predict blood pressure using age, sex, weight

High-Dimensions

$n \ll p$

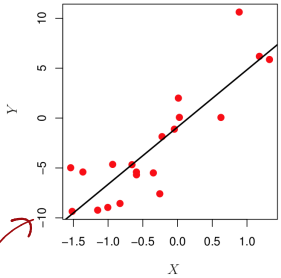
- Classical approaches not appropriate because overfitting!

→ predict blood pressure on genetic data

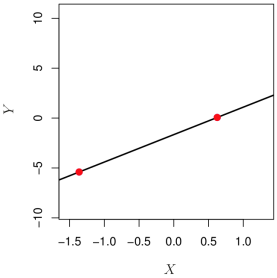
What goes wrong?

$p = 1$

100? $= n \gg p$



$n \approx p$



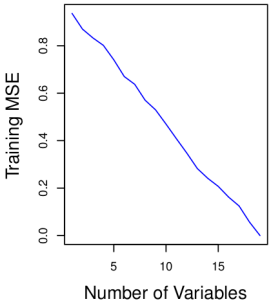
fits the training set exactly

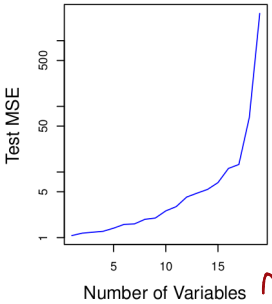
- but - my test error will be awful

Not perfect fit

- but - will do better w/ test error

More issues with least squares on big p





- $n = 20$
- Regression on $p = 1, \cdots, 20$
- Y completely unrelated to variables

training goes down no matter what

Test MSE gets higher
Leads to variance of coeff estimates

Always do CV

The answer to dealing with big p

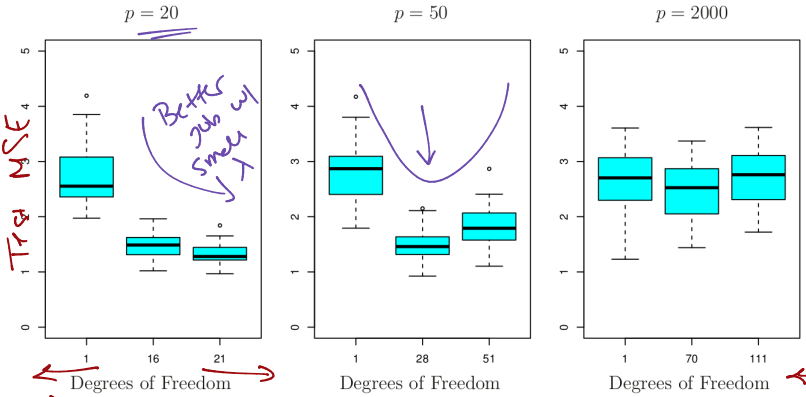
Be less flexible!

↳ Make p smaller!

- Subset selection (literally make p smaller)
- Shrinkage versions
  - ↳ Ridge
  - ↳ Lasso
- Dimensionality reduction
  - ↳ PCA + Regression PCR
  - ↳ PLS

Example with Lasso

Increasing p



- $n = 100$
- Boxplots = Test MSE
- DF = # non-zero coeffs

Key points

1. regularization or shrinkage play key role in high dim problems
2. Pay attention to tuning parameters (λ, M) Necessary for good models in high dimensions
3. The test error tends to increase as the dimensionality increases unless the added the additional predictors are actually associated to the response.

Curse of dimensionality

Phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings.

- You might think
  more variables =
  better fit

- But -

  adding unnecessary variables
  makes the Test MSE worse

Interpretation in high dimensions

Multi-collinearity: the concept that the variables in a regression might be correlated with each other

High dim: any variable might be itself a linear combo of other variables.

- Never know which of these is actually predictor of outcome
- Many possible best models since might use any of the correlated variables.

Reporting errors in high dimensions

$$\boxed{\text{Test MSE}}$$

$$\Uparrow$$

k-fold CV

Next time

10	M	Oct 3	Leave one out CV	5.1.1, 5.1.2	
11	W	Oct 5	k-fold CV	5.1.3	
12	F	Oct 7	More k-fold CV,	5.1.4-5	
13	M	Oct 10	k-fold CV for classification	5.1.5	HW #4 Due
14	W	Oct 12	Resampling methods: Bootstrap	5.2	
15	F	Oct 14	Subset selection	6.1	
16	M	Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
17	W	Oct 19	Shrinkage: Lasso	6.2.2	
18	F	Oct 21	[No class, Dr Munch out of town]		
	M	Oct 24	No class - Fall break		
19	W	Oct 26	Dimension Reduction	6.3	
20	F	Oct 28	More dimension reduction; High dimensions	6.4	HW #6 Due
	M	Oct 31	Review		
	W	Nov 2	***Midterm #2***		